

SHAMIL IRFAN F

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SUMMARY

Final-year Computer Science and Engineering student with a passion to build practical software that makes everyday tasks easier. Has experience with AI and computer vision systems, including embedded hardware. Strong foundation in programming, and Linux-based systems, with practical experience integrating AI into real-time applications and IoT devices.

TECHNICAL SKILLS

AI & Computer Vision: DeepFace, OpenCV, Ollama, LM Studio

Languages: Python, C, Java, Bash, Lua

Backend & Databases: Flask, REST APIs, MySQL, MongoDB

Tools & Platforms: Docker, Git, Linux, GCP

INTERESTS

Self Hosting

System Customization

Electronics

Astrophysics

PROJECTS

Adaptive Facial Recognition Framework for Secure Authentication — *Ongoing* 2026

Developing an academic AI research project combining anti-spoofing detection and occlusion-aware recognition to secure facial authentication against presentation attacks and partial occlusions (masks, glasses). Completed the zeroth review presentation; currently developing the semantic segmentation model.

AI-Enabled IoT-Based Real-Time Home Monitoring System — *Ongoing* 2026

Building an edge AI system on a Raspberry Pi that runs a computer vision model to detect movement and presence in real time, paired with a DHT22 sensor for temperature and humidity tracking. Streams live status to a web dashboard, with configurable threshold-based alerts for anomalies.

ClassSnap — Facial Recognition Attendance System 2025

Built the Flask REST API backend and integrated DeepFace for real-time facial recognition, storing attendance records in MongoDB. Collaborated with teammates on a React-based analytics dashboard and integrated a Raspberry Pi camera module, achieving 90%+ recognition accuracy.

YouTube Playlist Categorizer 2025

Built a Python tool that uses the YouTube Data API with a local LLM backend (LM Studio) to auto-sort playlist videos via structured JSON outputs. Uses a YAML-driven category configuration and auto resume sorting on failure.

Dynamic Numpad Macro System (MacroPad) 2024

Built a macro system with AutoHotKey that turns a numpad into a programmable macro pad, using the Interception driver to capture keyboard input at driver level and launch apps and programs.

HomeLab — Self-Hosted Personal Infrastructure 2024 – Present

Designed and maintained a Linux-based homelab across a Raspberry Pi and laptop using Docker, running a local AI stack (Ollama, LM Studio) for offline inference alongside self-hosted services: Jellyfin (media), Nextcloud and Syncthing (storage), Pi-hole (DNS), Tailscale and Nginx (remote access).

EDUCATION

B.E. Computer Science and Engineering Expected 2027

Coimbatore Institute of Engineering and Technology, Coimbatore

CGPA 8.31

Higher Secondary Education May 2022

Campion Anglo Indian Higher Secondary School, Tiruchirappalli

77%

Secondary School Education Mar 2020

Campion Anglo Indian Higher Secondary School, Tiruchirappalli

84%

CERTIFICATIONS

- NPTEL — Cloud Computing (Oct 2024), Modern Programming in C++ (Apr 2024)